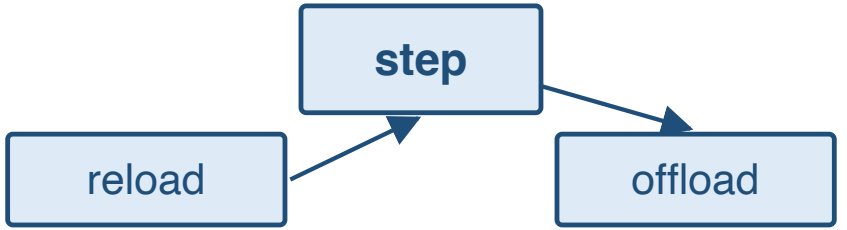
state offload
step on GPU

forward + backward of all micro-batches

HBM free of state



CPU optimizer
step on host

forward + backward of all micro-batches

host CPU does the math

host side

state = exp_avg + exp_avg_sq + FP32 master weights